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Introduction:

The primary aim of the report is to investigate the time-domain(transient) and frequency-domain (Steady-state) behavior of RLC circuits, and to design signal filters.

The study focuses on verifying theoretical damping conditions via simulation tools such as Multisim.

Scope: The Report is divided into four main sections:

1. Section 1: Series RLC Circuit (Transient & Resonance Analysis)
2. Section 2: Parallel RLC Circuit (Transient & Resonance Analysis)
3. Section 3: Passive Filters (Designing RC and RL filters (
4. Section 4: Active Filter (Op-Amp Design)

Section 1: Series RLC Circuit (Transient & Resonance Analysis)

Part A: Design & Transient:

1.1 Design Rationale and Components Selection:

To investigate the Transient behavior of a Second-order circuit, which is characterized by different equations, and consist of resistors and equivalent two energy storing elements (Alexander, C.K. and Sadiku, 2021, ch.8). A series RLC circuit was designed to operate at a specific natural resonant frequency (ω).

For this analysis, an angular frequency of **500 rad/s** was selected. This value was chosen to ensure that the critical resistance would be a standard integer value, mitigating rounding errors and theoretical validation.

The fixed reactive components were chosen as follows:

- **Inductance (L):** 1 H
- **Capacitance (C):** 4 μ F

Using the resonance formula $\omega = \frac{1}{\sqrt{LC}}$, these values:

$$\omega = \frac{1}{\sqrt{1 \times 4 \times 10^{-6}}} = 500 \text{ rad/s}$$

To demonstrate the three damping conditions (Overdamped, critically damped, and Underdamped), the resistance (R) varied while keeping L and C constant. The Theoretical critical resistance was calculated as:

$$R = 2\sqrt{\frac{L}{C}} = 2\sqrt{\frac{1}{4\mu}} = 1000 \Omega$$

Table 1 below (summarizes) the resistor value chosen for the analysis:

Damping State	Resistor Value	Condition
Overdamped	4000 Ω	$\alpha > \omega$
Critically damped	1000 Ω	$\alpha = \omega$
Underdamped	200 Ω	$\alpha < \omega$

Table [1]

1.2 Theoretical Framework:

The behavior of an RLC circuit is determined by comparing the Neper Frequency (α) and the Resonant Angular Frequency (ω).

1- **Neper Frequency** : $\alpha = \frac{R}{2L}$

2- **Resonant Frequency** : $\omega = \frac{1}{\sqrt{LC}}$

1.3 Simulation Setup:

The Circuit was implemented in Multisim using an Ideal DC Voltage Source (12V) and a time-controlled switch to create a step at $t = 0$

Initial Conditions: were set to user-defined, to simulate the capacitor charging from an uncharged state. The starting time was set to Zero, and the end time was set to 1.5 seconds.

Time Configuration: The Maximum Time Step (TMAX) was manually set to 0.1ms, ensuring high resolution.

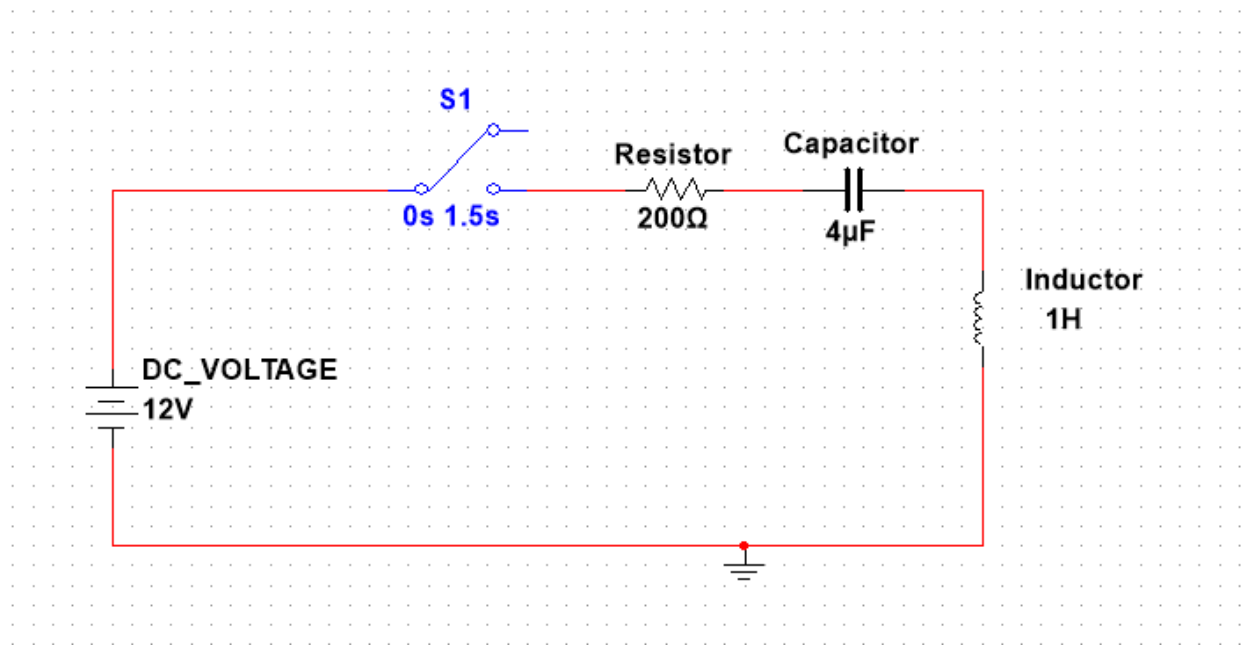


Figure 1: Multisim implementation of the series RLC circuit

1.4 Simulation & Results:

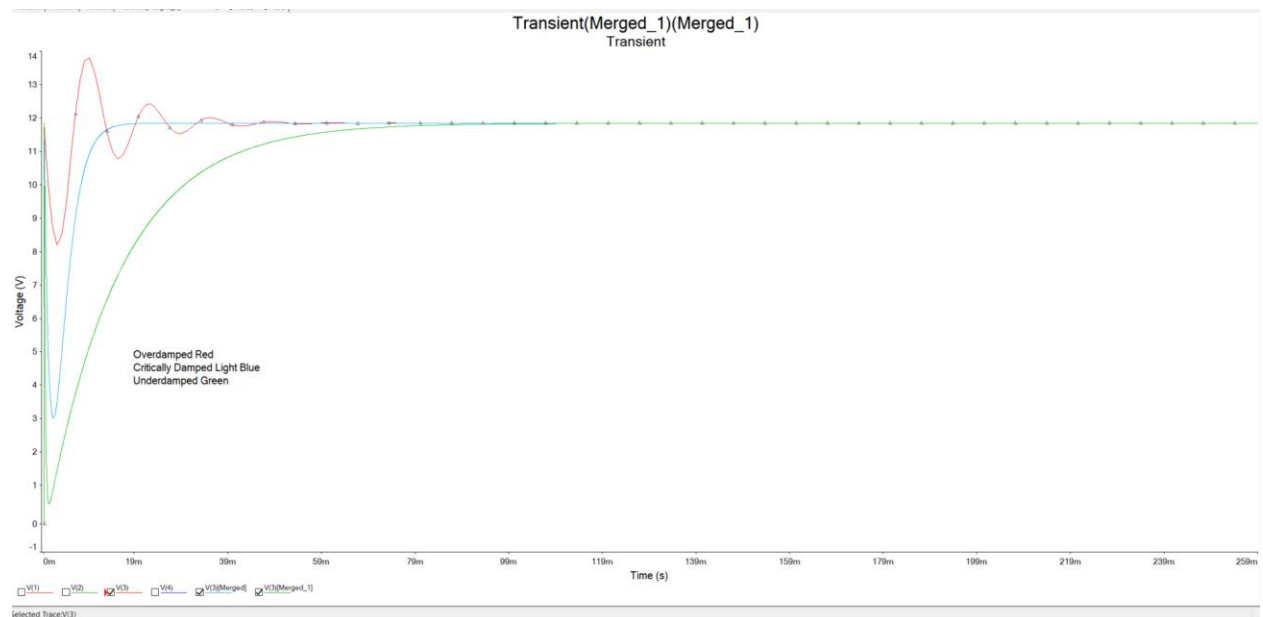


Figure 2: Damping conditions of Series RLC

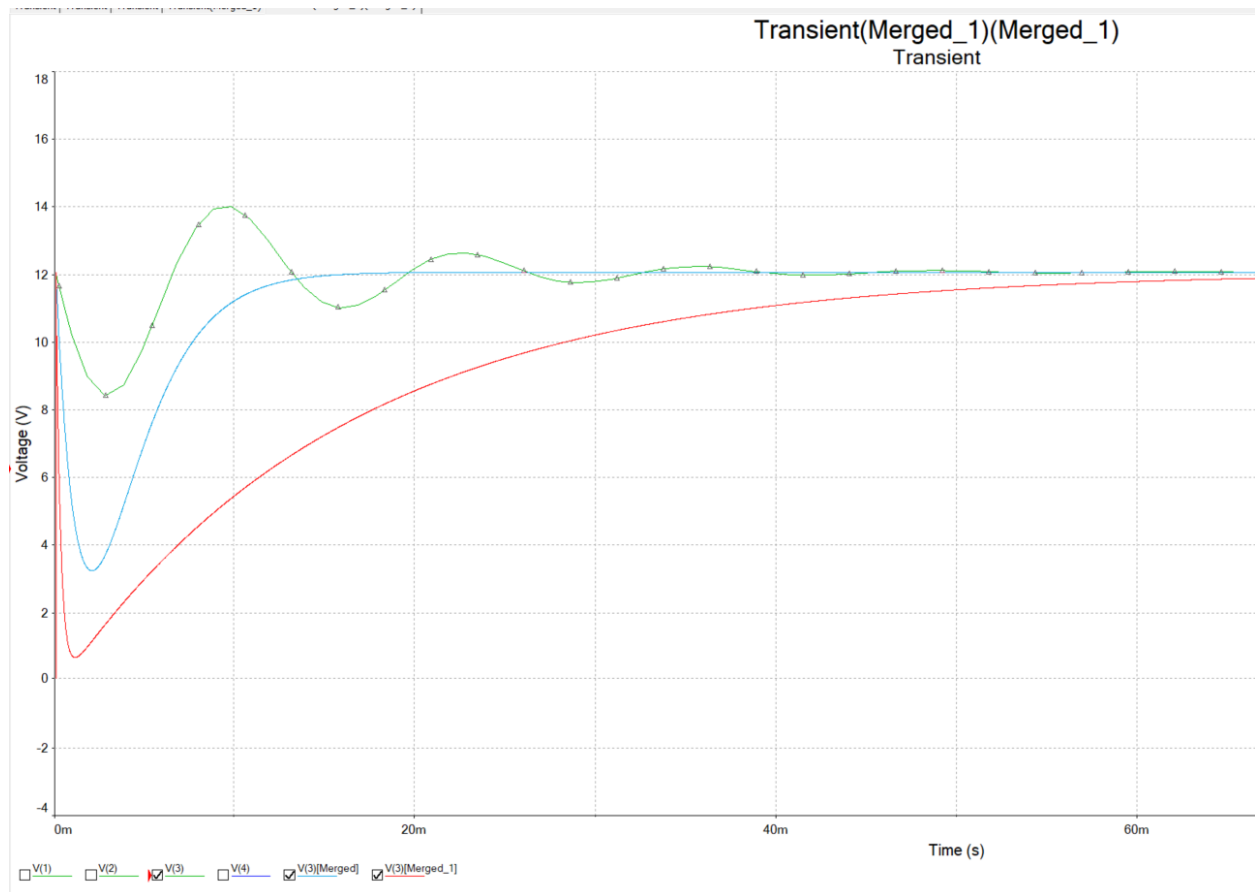


Figure 3: Damping conditions of Series RLC showing 12 V Overshoot

Analysis of Results:

The Transient response of the voltage across the capacitor was simulated for the three resistances to validate the theoretical damping conditions.

1- Underdamped Response ($R = 200\Omega$): As observed the green curve, the low resistance leads to a damping ratio of Neper frequency(α) greater than the Angular frequency (ω). The graph overshoots the 12V source and oscillates before settling down.

2- Critically damped ($R = 500\Omega$): The graph illustrates the critical damping as the Neper frequency(α) is equal to the Angular frequency (ω). The voltage rises to 12V without overshooting, and no oscillation occurred.

3- Overdamped ($R = 1000\Omega$): The Blue curve shows the effect of high resistance, leading to the Neper frequency(α) less than the Angular frequency (ω). This indicates that damping effectively eliminates oscillation compared to other cases.

Part B Resonance:

1.5 Design Rationale and Components Selection:

To investigate the frequency response characteristics of the Series RLC circuit, a specific resonant frequency (f_0) was chosen to demonstrate circuit response.

A frequency of 2 kHz (2000 Hz) was selected for the circuit design. This frequency lies within standard audio Bandwidth, making the analysis relevant to the real-world application like audio equalizers.

The component values were calculated based on a fixed capacitor value:

- **Capacitance (C):** 100 nF (0.1 μ F)
- **Inductor (L):** To achieve a resonance at 2kHz, the inductance was calculated using the resonance formula:

$$L = \frac{1}{(2\pi f_0)^2 C} = \frac{1}{(2\pi \times 2000)^2 \times 100 \times 10^{-9}} = 63.3mH$$

Resistor (R): A resistor with a resistance of 50 Ω was chosen. In a series resonant circuit, decrementing resistance increments the Quality Factor (Q), resulting in a sharper and more defined current peak for bandwidth measurement.

1.6 Theoretical Framework:

The resonance in a Series RLC circuit occurs when the inductive Reactance (X_L) and Capacitive Reactance (X_C) are equally in magnitude, but opposite in phase. At this frequency, both reactance cancels each other, forming the circuit purely resistive.

1- Resonant Frequency (f_0):

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{0.633 \times 100 \times 10^{-9}}} = 2001Hz$$

- 2- **Quality Factor (Q):** The Quality Factor determines the selectivity of the circuit. A higher Q value indicates a narrower bandwidth and better frequency rejection.

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{1}{50} \sqrt{\frac{0.0633}{100 \times 10^{-9}}} = 15.9$$

- 3- **Bandwidth (BW):** The Bandwidth represents the range of frequencies.

$$BW = \frac{f_0}{Q} = \frac{2000}{15.9} = 126\text{Hz}$$

1.7 Simulation Setup:

The frequency response was simulated in Multisim using the AC Sweep analysis function.

Source: An AC voltage Source was used and configured to 1 V_{pk} for the calculation.

Sweep parameters: The AC Sweep was configured to start at and stop at with decade type sweep. A Logarithmic scale was set to ensure the graph analysis centering the peak.

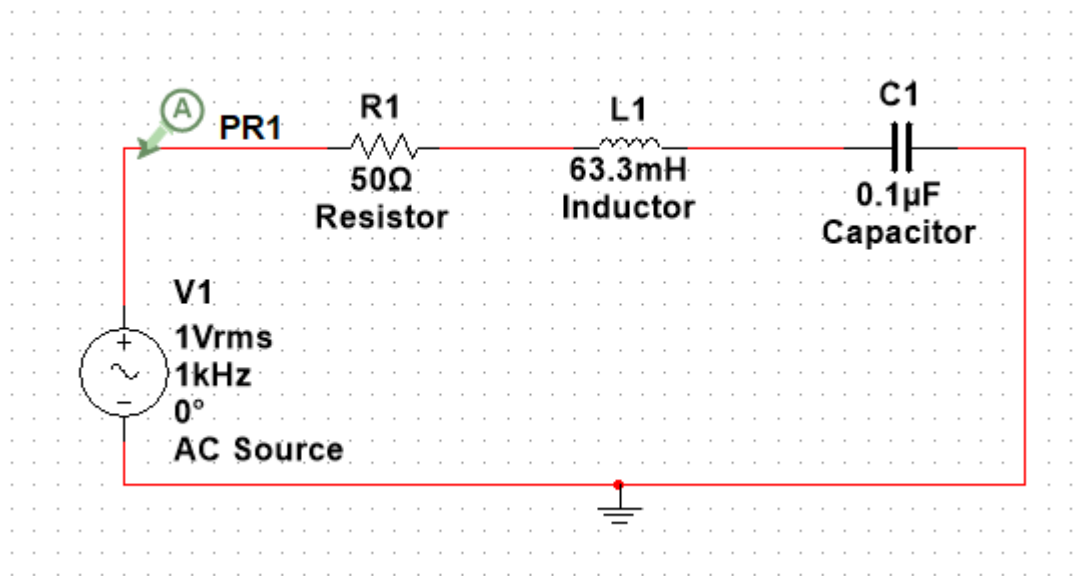


Figure 4: Series Resonance RLC Schematic

1.8 Simulation & Results:

The AC simulation results are presented in Figure below.

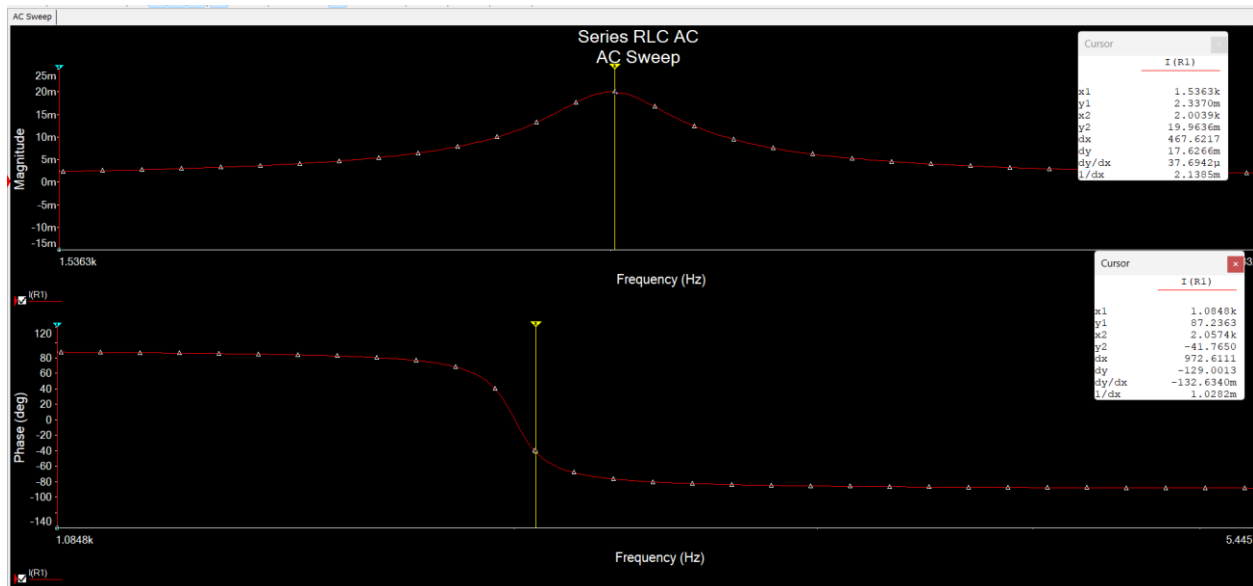


Figure 5: Series Resonance RLC Graph

Analysis of Results:

1- Resonant Peak: as observed in the graph, the current reaches the maximum value at 2.0kHz. At this frequency, the circuit impedance is at its minimum, allowing the maximum flow of current from the source.

2- Selectivity: The Peak is narrow, verifying the high-quality factor achieved by the 50Ω resistor. Frequencies (significantly) below or above 2kHz are attenuated rapidly.

3- Phase Behavior: Below the resonant frequency, the circuit impedance is dominated by the capacitor; above the resonance frequency, it is dominated by the inductor.

Section 2: Parallel RLC Circuit (Transient & Resonance Analysis)

Part C: Design & Transient:

2.1 Design Rationale and Circuit Modification:

To analyze the parallel RLC response, the same reactive components ($L = 1\text{H}$, $C = 4\mu\text{F}$) were retained to maintain the natural resonant frequency of $\omega = 500\text{ rad/s}$

However, simulating a parallel circuit with an ideal DC voltage source represents a theoretical challenge. An ideal voltage source has zero internal resistance, which would constrain the voltage across the branches to a fixed value, preventing natural response. A series resistor was placed between the switch and the DC voltage source. This resistor limits the current, allowing voltage division across parallel components.

2.2 Theoretical Framework:

In a parallel RLC circuit, the damping conditions are inversely related to the resistance due to that damping factor is defined as:

$$\alpha = \frac{1}{2RC}$$

$$\frac{1}{2Rc} = \frac{1}{\sqrt{Lc}} = \frac{1}{2} \sqrt{\frac{L}{c}}$$

This derivation shows that high resistance leads to underdamping and low resistance leads to overdamping.

Damping State	Resistor Value	Condition
Overdamped	100 Ω	$\alpha > \omega$
Critically damped	250 Ω	$\alpha = \omega$
Underdamped	1000 Ω	$\alpha < \omega$

Table [2]

2.3 Simulation Setup:

The circuit was designed in Multisim with parallel branches (C|R|L) connected to the source via 1K Ω series resistor.

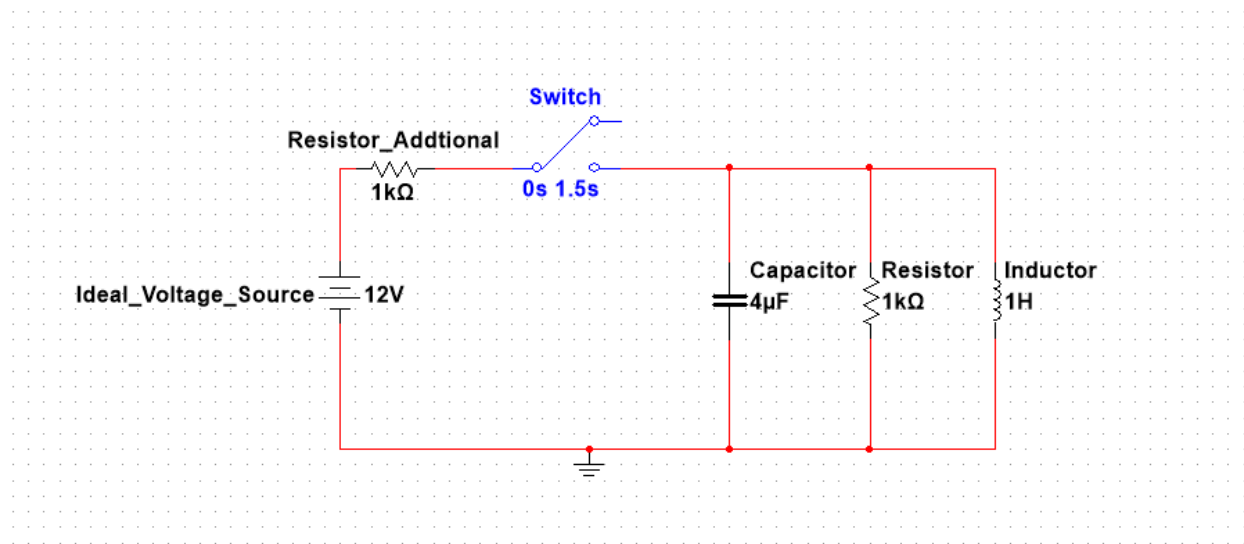


Figure 6: Parallel RLC Schematic

Initial conditions: were set to user defined, to simulate the capacitor charging from an uncharged state. Starting time was set to Zero and end time was set to 1.5 seconds.

2.4 Simulation Results and Interpretation:

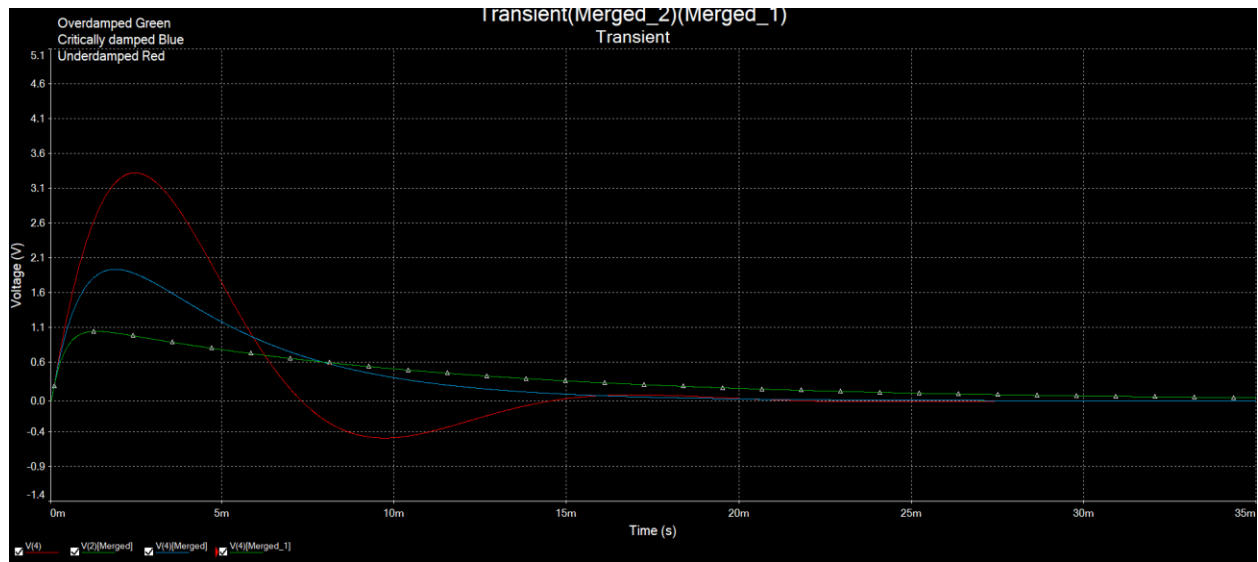


Figure 7: Merged Graph (Overdamped (Green), critically damped (Blue), underdamped (Red))

Analysis of Results:

The parallel RLC step response differs from the Series RLC steps response in two keys ways: the effect of the resistance and the final steady state-value.

1- Underdamped Response ($R = 1000\Omega$): A parallel high resistance prevents the energy from escaping; the graphs show the voltage peaks and oscillate before decaying.

2- Critically damped ($R = 250\Omega$): The graph illustrates the critical damping as the low parallel low resistance shows the voltage rising and then decaying without any oscillations.

3- Overdamped ($R=100\Omega$): The overdamped state was simulated using a low parallel resistance of 100Ω . The voltage rises smoothly and then decays back without any oscillations.

Part D Resonance:

2.5 Parallel Resonance Design (2 kHz):

To compare the frequency response with the series assembly, the parallel circuit was assembled using the same reactive components and calculated in section 1 part C ($L = 63.3\text{mH}$, $C = 100\text{nF}$).

Design Change: Unlike a series circuit, where decreasing resistance increases the selectivity Q , the parallel circuit requires a high resistance to maintain the same Q -factor ($Q = R / \omega L$).

Main Resistor (R): A load of $10\text{k}\Omega$ was selected. (As the 50Ω resistor would damp the circuit)

Series Resistor (R): A $1\text{k}\Omega$ resistor was placed to model source impedance.

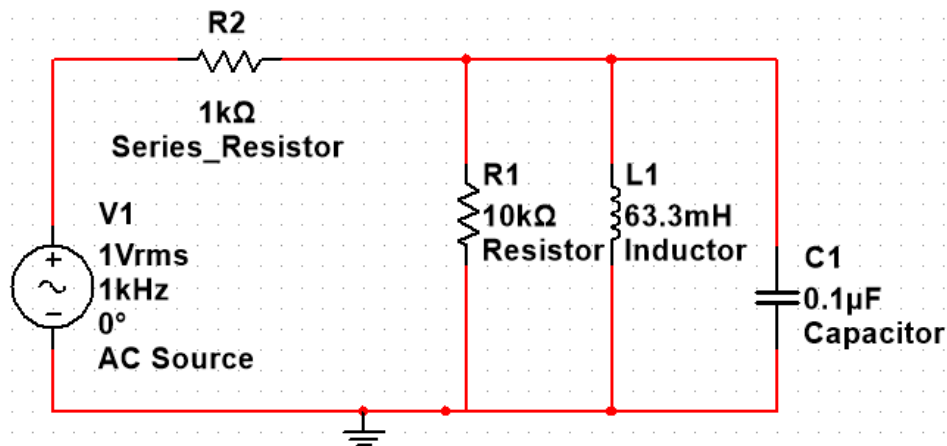


Figure 8: Parallel RLC Resonance Schematic

2.6 Theoretical Framework:

1- Resonant Frequency (f_0):

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{0.633 \times 100 \times 10^{-9}}} = 2001 \text{ Hz}$$

Quality Factor (Q): For a parallel circuit, $Q = \frac{R_p}{x_L}$

$$Q = \frac{10000}{2\pi(2000)(0.0633)} = 12.6$$

2.7 Simulation Results:

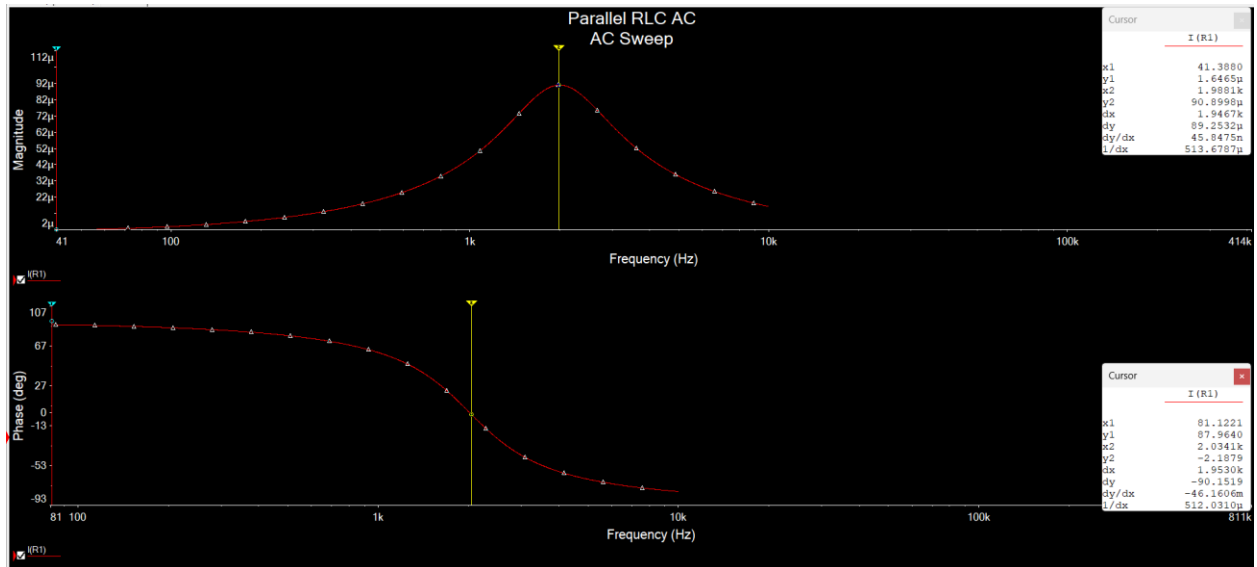


Figure 9 Parallel RLC Resonance Graph

Analysis of Results:

1- Voltage Peak: The simulation confirms a maximum value at 2.0kHz. At resonance, the inductive and capacitance currents are equal, canceling each other.

2- Comparison with Series:

Series Resonance: Maximum current (Minimum impedance)

Parallel Resonance: Maximum voltage (Minimum impedance). Both circuits were successfully tuned to 2kHz, verifying the duality of the RLC circuit.

Section 3: Passive Filters

Part A: RC Low-Pass Filter

3.1 Design and Theoretical operation:

A Low-Pass Filter circuit is a Circuit that passes low frequency signal while maintaining the same amplitude. The circuit uses Capacitor Reactance (X_C) to decrease as frequency increases.

The RC circuit is assembled to pass low frequency signals without attenuating frequencies above the cutoff frequency (1kHz). The circuit functions as a voltage divider but depends on frequency.

Theoretical operation: The output voltage is determined by the ratio of the capacitive reactance (x_c) to the total circuit Impedance(Z) :

$$v_0 = \frac{x_c}{Z} v_{in} = \frac{x_L}{\sqrt{R^2 + [x_L - x_c]^2}}$$

Capacitive Reactance: $X_C = \frac{1}{2\pi fC}$. There is an inverse the relation between frequency and reactance.

Low Frequencies: As $f \rightarrow 0, x_C \rightarrow \infty$. The Capacitor acts as an open circuit consequently, as the input voltage is equal to the output voltage, while the amplitude is maintained.

High Frequencies: $f \rightarrow 0, x_C \rightarrow \infty$. The Capacitor acts as a short circuit to the ground, leading the output voltage to drop.

3.2 Simulation Setup & Schematic

The circuit was assembled in Multisim using 1.6k Ω resistor and a 1 μ F capacitor to target cutoff 1kHz.

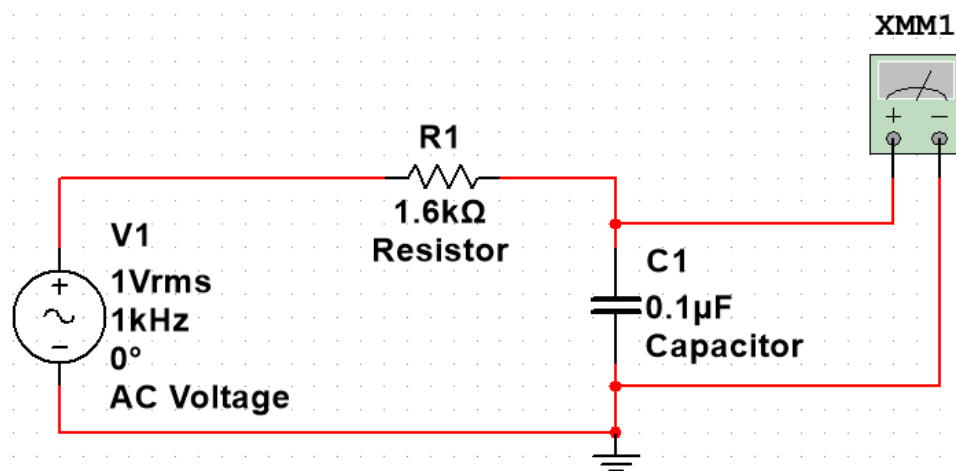


Figure 10: RC Low Pass Filter (Linear scale)

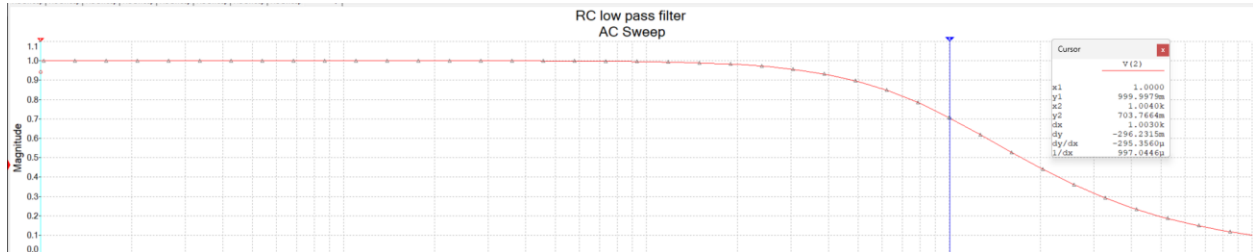


Figure 11: RC Low Pass Filter (Linear scale)

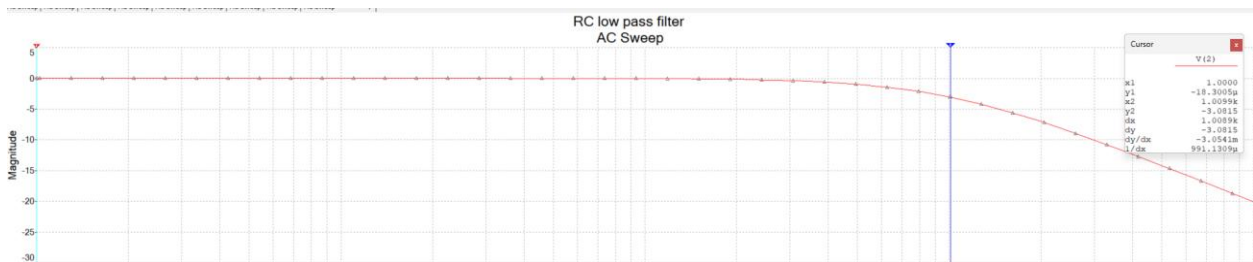


Figure 12: RC Low Pass Filter (Decibel scale)

3.3 Analysis of Result:

The simulation verifies that the theoretical cutoff frequency (f_c) of 1kHz.

Linear Scale Analysis Fig (11): in the linear voltage plot, the output starts at 1.0V (0) dB for low frequency. At a cutoff frequency of 1.00kHz, the voltage drops 707mv (or 0.707). This aligns with the theoretical calculation.

Decibel Scale Analysis Fig (12): As observed in the cursor, at a frequency of 1.00kHz (x_2), the magnitude drops to -3.01 dB (y_2). This confirms the definition of the cutoff frequency.

Part B: Low-Pass RL Circuit

3.4 Design and Theoretical operation:

The RL Low-Pass Filter uses an Inductor and a Resistor to block high-frequency signals. Unlike the capacitor which blocks DC, the inductor acts as a short circuit to DC and low frequencies, allowing them to pass.

Theoretical Operation: The circuit forms a voltage divider where the output is taken across a resistive element:

$$v_0 = \frac{R}{Z} v_{in} = \frac{R}{\sqrt{R^2 + x_L^2}}$$

Relationship between Frequency and Output: Since the inductive reactance is directly proportional to frequency ($x_L = 2\pi fL$):

Low Frequencies: As $f \rightarrow 0$, $x_L \rightarrow 0$. The Inductive reactance () is negligible, the inductor acts as a short circuit allowing full current flow across the resistor.

High Frequencies: $f \rightarrow \infty$, $x_L \rightarrow \infty$. Inductive Reactance becomes large. The inductor prevents the current flow from reaching the output resistor.

3.5 Simulation Setup & Schematic:

To maintain the same cutoff frequency the circuit was designed to have :

Resistor(R): 1k Ω

Inductor(L) : The required inductance were calculated as :

$$L = \frac{R}{2\pi f_c} = \frac{1000}{2\pi(1000)} \approx 159mH$$

Circuit Schematic:

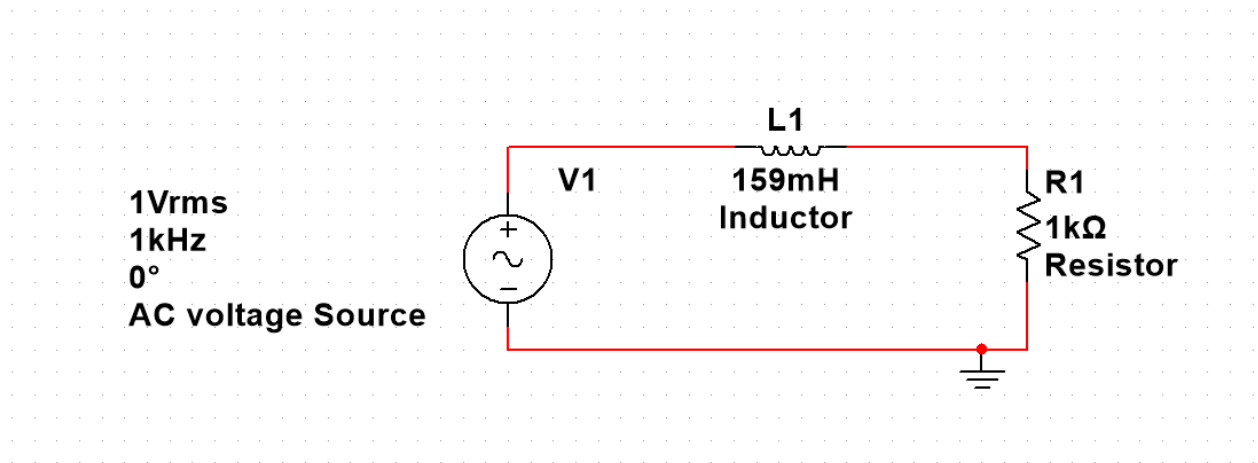


Figure 13: RL Filter Schematic

3.7 Simulation Results:

The AC Sweep demonstrates the attenuation of high frequencies.

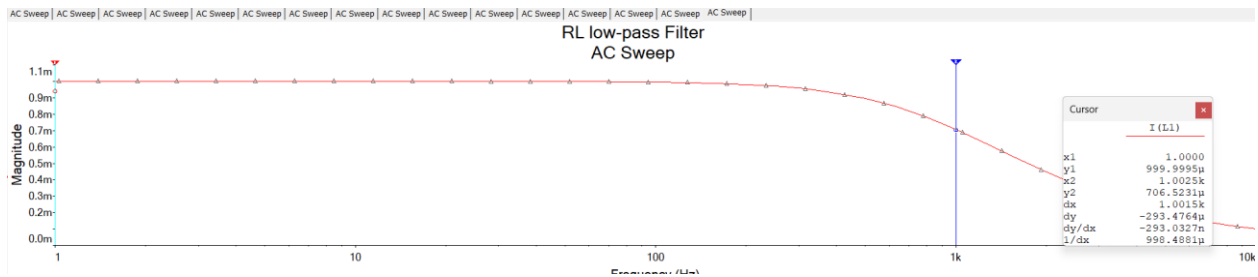


Figure 14: RL Low Pass Filter (Linear Scale)

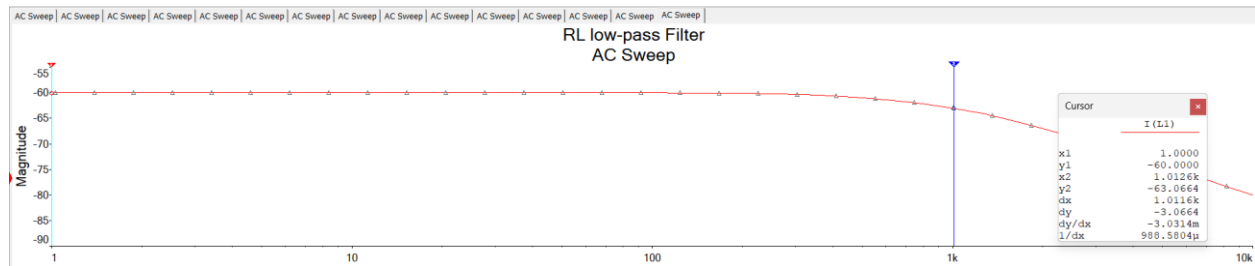


Figure 15: RL Low Pass Filter (Decibel Scale)

Analysis of Results:

The Simulation validates the RL Low-Pass characteristics:

Passband: At low frequencies the curve is flat 0 dB ($V_{out} = V_{in}$) confirming that the inductor offers opposition of low frequencies.

Cutoff Frequency: The curve crosses the -3dB point or (0.707) at exactly 1.00kHz.

Part C: CR High-Pass Filter

3.8 Design Rational

The CR High-Pass Filter is constructed by placing the Capacitor in series with the Voltage source and parallel with a resistor.

Theoretical Operation: The output voltage is determined by the voltage divider rule between the Resistor (R) and the Capacitive Reactance (X_c)

$$v_0 = \frac{R}{Z} v_{in} = \frac{R}{\sqrt{R^2 + x_c^2}}$$

Frequency Dependence logic: Circuit behavior is determined by the inverse relationship between frequency and reactance ($x_c = \frac{1}{2\pi fC}$):

Low Frequencies: As the frequency increases, X_c decreases. The Capacitor acts as a short circuit, allowing the input voltage to appear across the resistor (V_{out}).

High Frequencies: As the frequency increases, X_c decreases. The Capacitor acts as a short circuit, allowing the input voltage to appear across the resistor (V_{out}).

Component selection:

To maintain a consistent Cutoff Frequency (f_c) of 1kHz, the components were calculated as follows:

Capacitor (C): Fixed at 100nF (0.1 μ F)

Resistor (R):

$$R = \frac{1}{2\pi f_c} = \frac{1}{2\pi(1000)(100 \times 10^{-9})} = 1591\Omega$$

Selected Value: A standard 1.6k Ω selected.

3.9 CR High-Pass Filter:

Multisim Schematic:

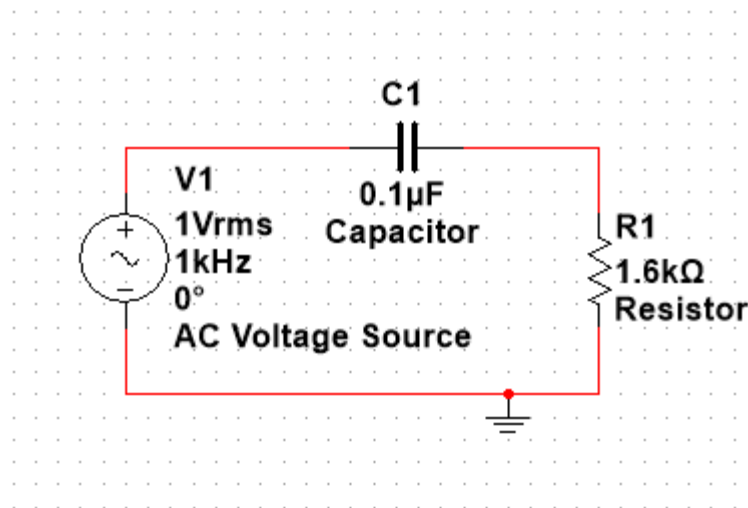


Figure 16: CR High-Pass Filter Schematic

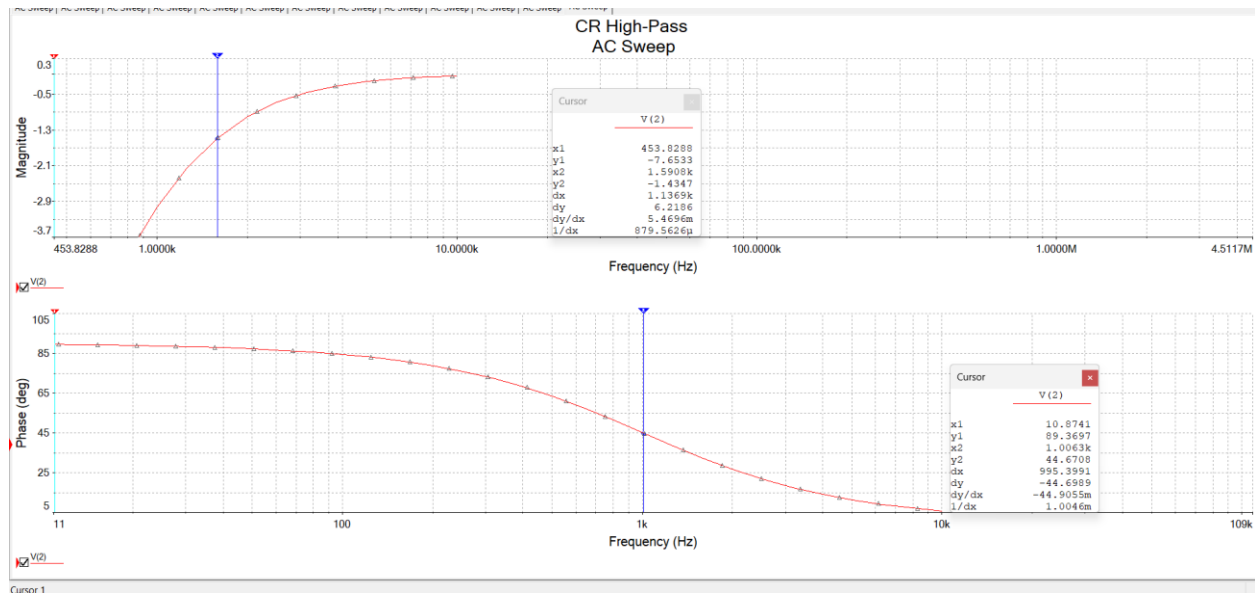


Figure 17: CR High-Pass Filter Graph

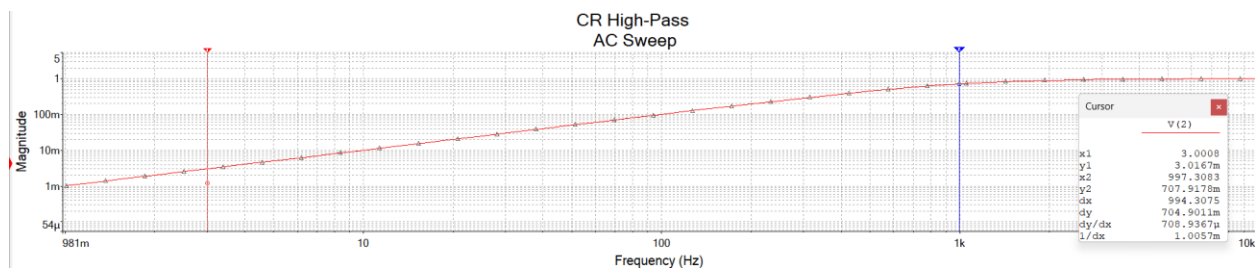


Figure 18: CR High-Pass Filter Graph (Linear)

Analysis of Results:

The simulation validates the High-Pass characteristics:

Stopband: At low frequency , the output is nearly zero because the series capacitor acts as an open circuit.

Cutoff Frequency: The Cursor on the graph confirms that at 1.00kHz , the magnitude is -3 dB (or 0.707), matching the theoretical design.

Passband: As frequency increases after 1kHz, the gain flattens out to 0 dB, allowing the signal to pass without attenuation.

Section 4: Active Filter (Op-Amp Design)

Part A: Active Low pass Filter

4.1 Design Rationale:

To improve the passive filters designed in Section 3, an Active Low-Pass Filter was designed using a UA741 Operational Amplifier. While passive filters are simple, they suffer from signal attenuation and are sensitive to the load impedance. The active design utilizes a non-inverting topology to buffer the signal and provide a voltage gain.

4.2 Design Calculations:

The design targets a cutoff Frequency of 1kHz and Gain of **(6dB)**.

1. Filter stage (Cutoff Frequency): Using the RC circuit at the non-inverting input:

Resistor(R1): 1.6kΩ

Capacitor(C1): 0.1μ

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi(1600)(100 \times 10^{-9})} = 995 \text{ Hz}$$

2. Amplifier Stage (Gain): Using the feedback resistors connected to the inverting input:

Feedback (R3): 10KΩ

Ground (R2): 10kΩ

$$A_v = 1 + \frac{R_3}{R_2} = 1 + \frac{10000}{10000} = 2$$

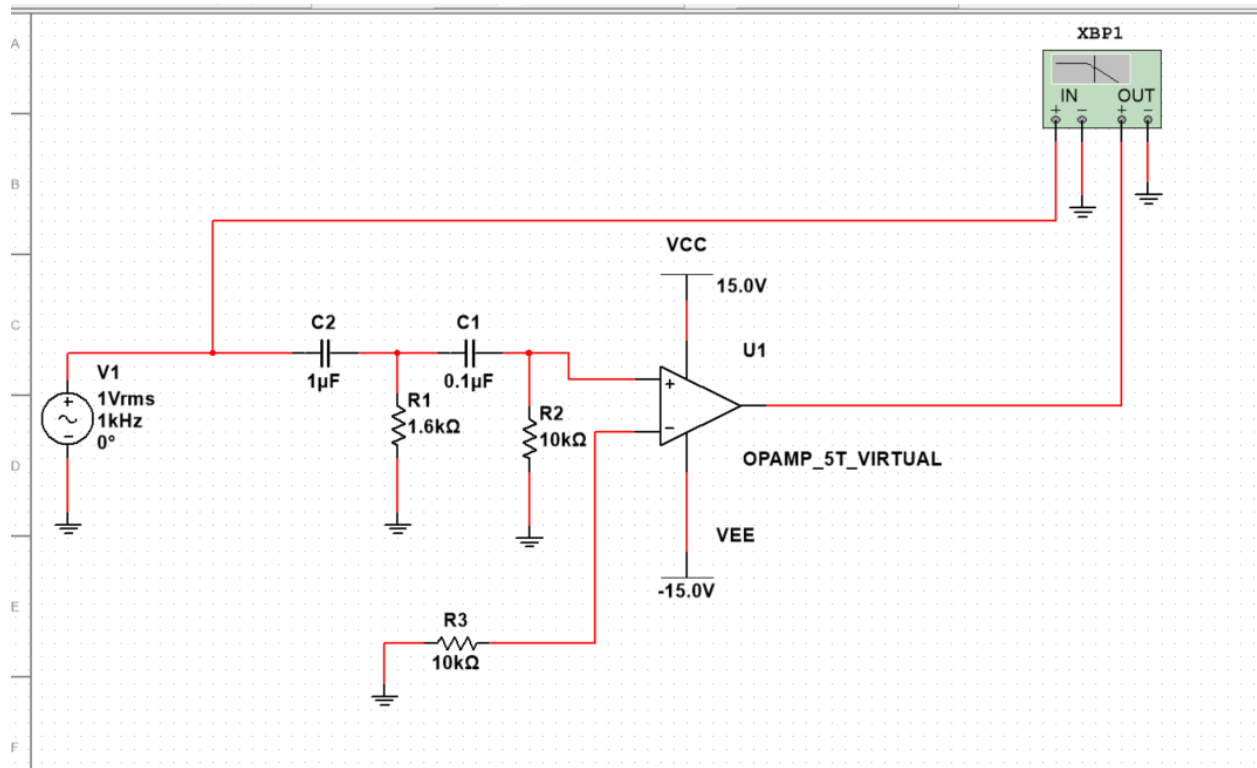


Figure 19 Active Low-Pass Filter

The circuit was implemented using dual DC rails (+15V and -15V). The input signal(1Vrms) passes through the RC filter into the non-inverting input, while the feedback network sets the gain.

4.3 Simulation Results

The AC Sweep analysis confirms the circuit operation.

Passband: The output voltage in the pass band is approximately 2V.

Filtering: The signal attenuates past 1kHz, confirming the low-pass behavior.

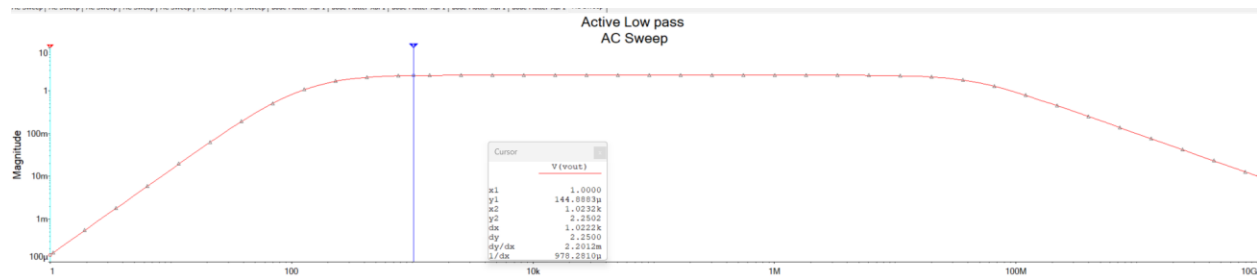


Figure 20 Graph of the Active Low-Pass Filter

Section 5: Real-World Applications

5.1 Series RLC Circuit: Radio and Frequency Tuning

The resonance behavior observed in section 1 (Part B) is the operating principle behind radio receivers. By adjusting the capacitance, the circuit's resonant frequency matches the carrier frequency of the radio station.

5.2 Passive Filters:

1- Power Supply Regulation

LC filters have a critical role in power supply regulation by effectively smoothing voltage fluctuations and minimizing ripple voltage. These filters are mainly assembled of inductors and capacitors. This combination filters efficiently unwanted frequency noise.

2- Audio Processing

LC filters play a critical role in audio processing by significantly influencing the frequency characteristics of sound signals. LC filters enhance or suppress ranges of sound, which is essential for achieving optimal audio quality.

5.3 Active Filters: Subwoofers

The active low-pass filter designed in section 4 is critical in a subwoofer system. In a home theater, the subwoofer channel must neglect all high frequency noise and boost weak signals from the receiver.

Conclusion:

1. RLC Transient Response: The simulations in Section 1 and 2 confirmed the damp conditions are controlled by the resistance. High resistance in a series circuit caused overdamping, while low resistance caused underdamping, matching theoretical predictions.
2. Resonance: The series RLC circuit demonstrated a sharp current peak at 2.0kHz, verifying the calculated resonant frequency formula $f_0 = \frac{1}{2\pi\sqrt{LC}}$
3. Filter Performance:
The Passive RC/CR Filters successfully set a cutoff frequency at 1kHz (-3 dB), effectively neglecting unwanted signals.

The Active Low-Pass Filter outperformed the passive design, maintaining 1kHz cutoff but also provided a Passband Gain of 6(dB). This sheds light on the advantage of active filters in preventing signal loss.

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